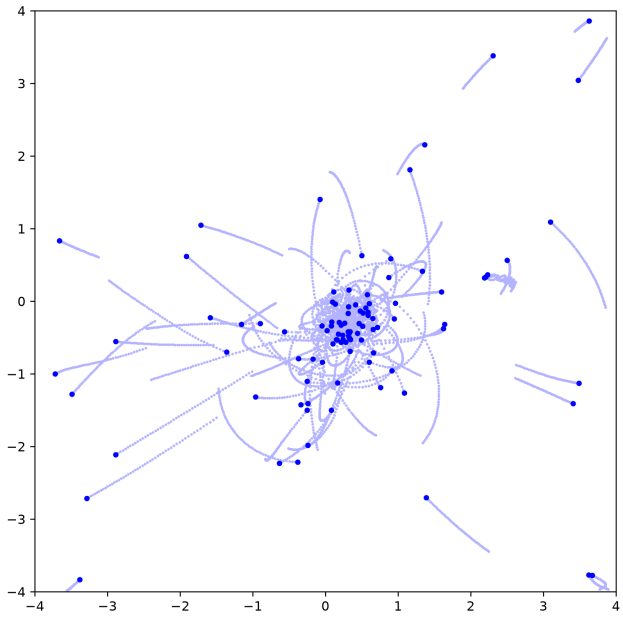
ECE 759 — Assignment 4 Pranav Mehrotra

GitHub Link: `https://github.com/pmehrotra3/repo759/HW04`

Task 1.

a) `numpy` and `matplotlib` installed.

b) `nbody.py` was run and the final plot, saved as `task1.png`, is shown below.



Task 2.

a) `task2.cpp` (C++ port of the N-body simulation, completed from the provided skeleton) present in `HW04` folder.

Task 3.

a) `task3.cpp` (OpenMP version of `task2.cpp`, taking `num_threads` as the third argument) present in `HW04` folder. `#pragma omp parallel for` is used on the outer particle loop of `getAcc`, the two half-step kicks, the drift and the board-limit check.

Task 4.

a) The scheduling policies are implemented in `task3.cpp`: all parallel loops use `schedule(runtime)`, and an optional fourth argument (`static`, `dynamic` or `guided`; default `static`) selects the policy, e.g. `./task3 1000 100 8 dynamic`.

b) The plots of the time taken by the N-body simulation versus `num_threads` for $N = 1000$ particles and simulation end time $= 100$, one for each scheduling policy (`task4.pdf`), are shown below.

